



OPEN Future flood susceptibility mapping under climate and land use change

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Floods are a significant natural hazard, causing severe damage. Understanding how climate change and land use and land cover (LULC) changes influence flood patterns is crucial for developing sustainable management strategies. This research aims to develop flood susceptibility maps considering the impacts of climate change and land use changes, providing insights into risks from urbanization and climate shifts. Three machine learning models—XGBoost, Random Forest (RF), and Support Vector Machine (SVM)—optimized with Particle Swarm Optimization, were applied to the flood-prone Kashkan watershed in Iran. Results showed that distance from the river, digital elevation model, precipitation, and LULC were the most influential factors. The RF model outperformed others in mapping flood-prone areas, with high-risk zones covering 20% (1908 km²) of the region, primarily in built-up areas. Land use projections for 2050, using the CA-MARKOV model, estimate built-up areas will expand to 859.3 km². Future precipitation patterns were examined using 8 selected general circulation models under the SSP126 and SSP585 scenarios. Analysis under the SSP585 scenario indicates a 1.9 km² rise in moderate flood areas, a 36.26 km² increase in high-risk zones, and a 21.94 km² decline in very low-risk areas, highlighting expansion of high and moderate flood risk areas.

Keywords Flood susceptibility mapping, Machine learning, Climate change, Land use change

Floods have long been a significant natural disaster impacting watersheds worldwide¹. In the past few decades, the influence of climate change has become progressively pronounced, with many studies predicting a substantial prolonged flood risk due to its effects^{2,3}. The Middle East and Northern Africa (MENA) region is especially susceptible to climate change, facing anticipated challenges such as rising temperatures, intensified heat waves, altered precipitation patterns leading to more severe droughts and floods, rising sea levels, and an increased risk of waterborne diseases^{4–7}. Given these challenges, it is crucial to study the impact of climate change on flood events within this region. Flood susceptibility mapping is vital for policymakers and authorities to develop effective emergency response plans. Accurately identifying flood-prone areas is essential for mitigating potential losses and damage^{8–10}. Changes in land use and land cover (LULC) are vital for flood susceptibility mapping. LULC, especially the growth of urban regions, have a profound impact on hydrological processes, including runoff patterns, peak flow attributes, water quality, and streamflow redirection^{11–13}. For instance, modifications to hillslopes for agricultural purposes—such as altering flow paths, velocities, and water storage—often lead to the formation of compacted soil layers. These layers enhance lateral flow while reducing vertical infiltration, consequently diminishing the filtering and buffering capacity of deeper soil layers¹⁴. Given these factors, it is crucial to integrate LULC changes when developing flood susceptibility maps for watersheds affected by climate change.

One approach to flood susceptibility mapping involves qualitative methods, such as the multi-criteria decision-making (MCDM) method, where weights for flood conditioning factors are assigned based on expert judgment. This method can lead to lower reliability and potentially inaccurate results across diverse study areas due to its reliance on subjective weight assignment and difficulty in capturing complex relationships among numerous flood-influencing factors^{15–17}. Consequently, the contributions of different factors to the final susceptibility map may be unclear, which can impair the model's ability to accurately predict future flood events. Another approach is numerical modeling, which, despite its utility, often suffers from errors and uncertainties and has limitations in replicating the complex physical processes of flows^{18–20}. In contrast, machine learning-based methods have demonstrated superior accuracy in flood susceptibility mapping compared to traditional models²¹. Machine learning techniques excel at capturing intricate patterns^{22,23} associated with events like floods and landslides¹⁵. Various machine learning methods have been employed for flood susceptibility mapping, including support vector regression (SVM)²⁴, convolutional neural networks (CNN)²⁵, and blended approaches utilizing dual-layer structures with components like SVM, logistic regression²⁶, K-nearest neighbor (KNN), and random forest (RF)^{25,27,28}, as well as artificial neural networks (ANN)²⁹. Furthermore, several studies have

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explored flood susceptibility mapping in the scope of climate change. For example, Madhuri et al.³⁰ utilized five machine learning algorithms to categorize locations as either flooded or non-flooded based on various climate change scenarios. Among these, XGBoost demonstrated superior performance. Avand et al.³¹ used Flexible Discriminant Analysis (FDA) and Artificial Neural Networks (ANN) to create probability flood maps for future climate change-induced floods. Gharakhanlou and Perez³² applied four machine learning models to predict future flood susceptibility under three climate change scenarios (RCP2.6, RCP4.5, and RCP8.5). Their research, conducted in two Canadian watersheds, found that the RF model had the best performance in generating flood susceptibility maps.

Existing knowledge

Flood susceptibility mapping has been a key research focus for decades, especially in the context of natural disasters and climate change. Various approaches have been developed to assess and predict flood risk, each with its strengths and limitations. Flood susceptibility mapping has traditionally relied on methods like multi-criteria decision-making (MCDM) and numerical modeling, but these often suffer from subjectivity and inaccuracies due to their inability to capture complex flood-related dynamics. In contrast, ML models such as SVM, RF, and Extreme Gradient Boosting (XGBoost) provide robust tools for flood susceptibility mapping due to their high accuracy and capability to handle complex datasets. These algorithms have demonstrated exceptional performance in previous studies related to natural hazard management, particularly in areas with diverse geographical features^{33,34}. Moreover, hyperparameter optimization is crucial for improving the performance, accuracy, and generalizability of machine learning models. By fine-tuning hyperparameters, models are better aligned with the specific characteristics of the dataset, which is especially important in complex tasks like flood susceptibility mapping³⁵. Using optimization algorithms such as Particle Swarm Optimization (PSO) helps efficiently explore the search space, avoid local optima, and achieve faster convergence to optimal solutions³⁶. This process improves predictive accuracy, enhances model robustness, and mitigates overfitting or underfitting, making predictions more reliable for environmental hazard management³⁷. Additionally, LULC changes, particularly urbanization and agriculture, significantly influence hydrological processes and flood risks, highlighting the need for their consideration in flood susceptibility mapping.

The current study

Although several studies in the literature integrate climate change with flood susceptibility mapping, the innovation of this study lies in the projection of future LULC and using it with the latest CMIP6 climate models to project future climate change scenarios. While climate change projections are frequently utilized, fewer studies consider both the impact of these projections and the dynamic changes in LULC, which can significantly influence flood vulnerability. This approach enhances the precision of flood predictions while offering insights into the impact of climate and anthropogenic changes on flood risks. Furthermore, while Middle East is home to low-income populations that suffer substantial losses from floods, there remains a considerable research gap regarding the region's flood vulnerability, particularly when considering the synergistic impacts of climate change and LULC changes. By addressing this gap, our study provides a novel approach that combines state-of-the-art climate projections and detailed LULC change analysis by using ML models integrated with a novel optimization method (beyond the conventional grid search methods used in many studies) to assess flood susceptibility in Middle East. Therefore, the objectives of this study are: (i) to assess the combined impact of LULC and climate change in one of Iran's most flood-prone sub-basins, known for its long history of flood events, and (ii) to develop flood susceptibility and probability maps for the future, utilizing CMIP6 projections under SSP126 and SSP585 scenarios, by applying three machine learning models integrated with PSO.

Materials and methods

The methodology employed in this study integrates multiple data sources and advanced machine learning techniques to map flood susceptibility. The dataset, consisting of 24 topographical, hydrological, and environmental factors, was gathered from reliable national and international sources. Factor selection was performed using the Information Gain Ratio method, prioritizing the most impactful predictors to enhance model accuracy. These factors were validated for multicollinearity using Variance Inflation Factor (VIF) analysis and Tolerance testing, ensuring the absence of significant collinearity. Machine learning models, including RF, SVM, and XGBoost, were optimized using PSO to improve predictive accuracy. In addition, future climate projections for precipitation patterns were derived from CMIP6 models under SSP126 and SSP585 scenarios to account for potential climatic variations. LULC changes were simulated using the CA-Markov method to capture spatial and temporal shifts in land development. This framework approach offers a comprehensive assessment of flood susceptibility under changing environmental and climatic conditions, ensuring a reliable mapping of current and future flood-prone areas. In this study, we employed Support Vector Classification (SVC), along with RF and XGBoost algorithms using RandomForestClassifier and XGBClassifier. All methods yield probabilistic outputs indicating the likelihood of a positive class ('flood'). These continuous outputs, ranging from 0 to 1, reflect confidence levels, ensuring robust and interpretable classification outcomes.

Traditional statistical methods like logistic regression and AHP fail to capture the nonlinear and multidimensional relationships inherent in flood susceptibility data, often overlooking complex interactions among factors such as soil type and land-use changes³⁸. In contrast, machine learning algorithms offer greater flexibility and precision for modeling these complexities. This study enhances the SVM, RF, and XGBoost models with PSO, improving both accuracy and efficiency. By conducting a comparative analysis of these algorithms with PSO for the Kashkan River Basin, the research highlights their strengths and adaptability in handling the basin's unique data characteristics. The 24 factors contributing to flood occurrence were selected based on the study area's characteristics and the availability of data, and these factors were mapped following the data

preparation steps (Figures S2, S3, and S4). In addition, each flood explanatory factor was mapped following the data preparation steps. To ensure consistency, all factors were processed to have a comparable spatial resolution of 30 m per pixel, in line with the spatial resolution of the DEM data (i.e., 30 m). This integrated methodology, which includes factor selection, machine learning model development, climate change downscaling, LULC simulation, and future flood susceptibility mapping (Fig. 1), provides a robust framework for evidence-based decision-making in flood risk management.

Study area

The Kashkan watershed, located between 47° 12' and 48° 59' E longitudes and 33° 8' to 34° 2' N latitudes (Fig. 2), is well-known for its flood events and its major rivers. Historically, these rivers have supported the development of numerous settlements along the waterways, fostering the creation of densely populated areas with high agricultural productivity. Spanning an area of 9560 square kilometers, the Kashkan River constitutes over 33% of Lorestan's total area. It stretches 374 km in length, with a gradient from 2400 m in the northern part of the province to 500 m near Poldokhtar city in the south, resulting in an average slope of 0.5%³⁴. The river has a mean yearly discharge of 52 m³/s, equating to a total annual discharge of 1.636 billion cubic meters. The Kashkan basin is highly prone to flooding and has experienced significant flood damage multiple times. On April 1, 2019, the main river recorded its highest flow of roughly 5232 m³/s at the watershed's outlet. This extreme

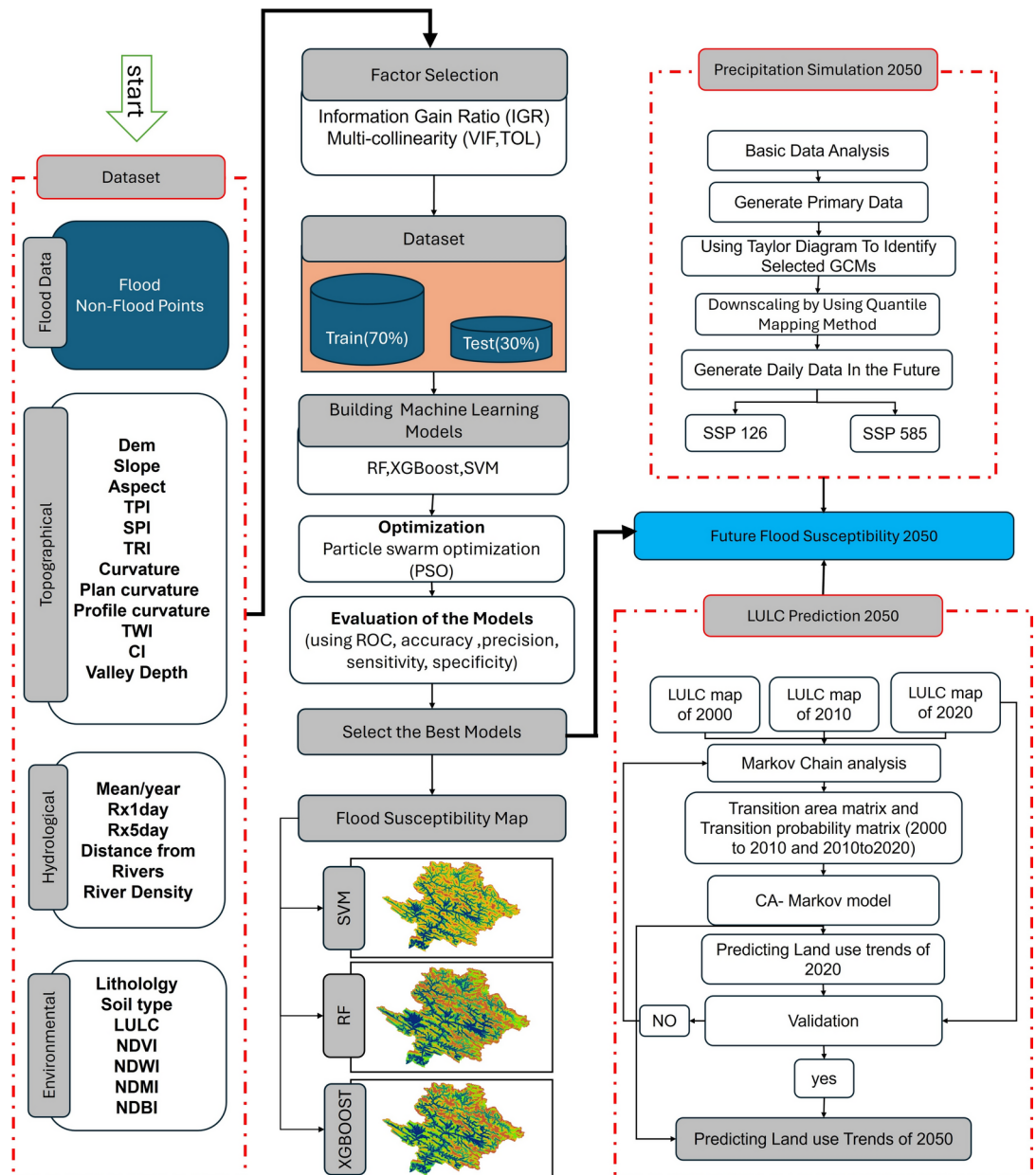


Fig. 1. The study's methodology, including the selection of key factors, application of machine learning models and steps for mapping current and future flood susceptibility in the Kashkan basin.

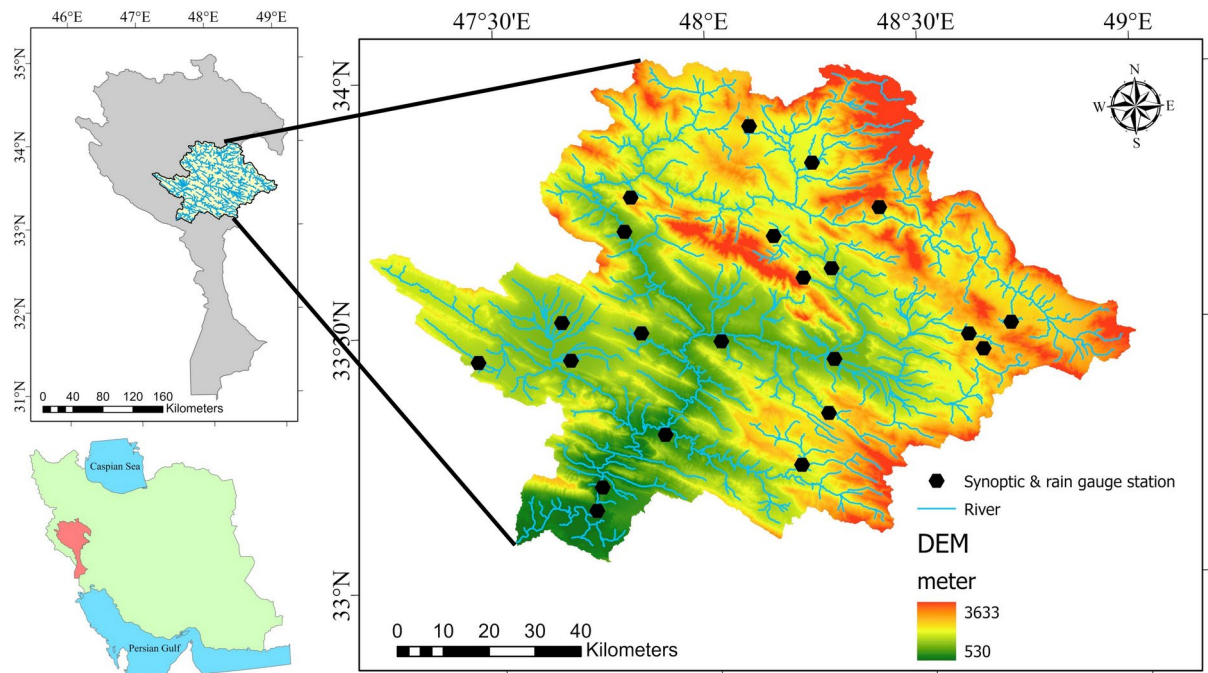


Fig. 2. Geographical map of the Kashkan basin, highlighting stations, river networks, and elevation changes [the map was generated by QGIS 2.18 (<https://qgis.org/>)].

event caused widespread damage to residential areas, roads, farmlands, and vital infrastructure, highlighting the critical need for efficient flood management strategies in the region. In addition to the climatic variability, the Kashkan River basin's hydrological regime is influenced by the region's complex topography and seasonal snowmelt from the Zagros Mountains. This contributes to substantial runoff during the spring and early summer months, exacerbating flood risks, particularly during periods of intense precipitation. Furthermore, the rapid urbanization in and around the Kashkan watershed has led to a significant reduction in natural flood mitigation systems, such as wetlands and riparian vegetation, which previously played a crucial role in floodwater retention and absorption. The interplay between land-use change, climate variability, and the basin's physiographic characteristics presents a challenging scenario for sustainable flood risk management in the region^{35,36}.

Flood inventory

Analyzing historical flood occurrences is crucial for predicting future floods, given the strong correlation between past and future events^{39–41}. Historical flood data were sourced from the Iran Ministry of Energy, the Iran Water and Power Resources Development Company (IWPCO), and the Iran Space Agency, and were validated using Sentinel-1 imagery. This dataset includes information on two major flood events that occurred in the region in 2013 and 2019. While flooded sites are generally well-documented, the determination of non-flooded locations is often inadequately explained. Some studies choose non-flooded sites based on stringent criteria, such as higher elevations and high drainage capacity, incorporating expert opinions to enhance model accuracy and convergence^{42–45}. To address this issue, this paper recommends a more objective approach by randomly selecting non-flooded sites distributed across historically non-flooded areas, thereby minimizing the impact of human judgment on model training⁴⁵. A total of 493 locations with historical flood occurrences were detected, and a total of 484 non-flood sites were randomly chosen using QGIS software (Figure S1). For model development and training, flood locations were coded as 1 and non-flood locations as 0. The total dataset of 977 points was segmented into a training set (70%) and a testing set (30%), as commonly practiced in susceptibility studies^{38,46}.

Topographical factors

Elevation, slope, aspect, topographic position index (TPI), stream power index (SPI), topographical roughness index (TRI), topographical wetness index (TWI), curvature, plan curvature, profile curvature, and convergence index (CI), along with valley depth, are considered topographical features derived from digital elevation models. These topographical features are calculated using the Spatial Analyst tool in QGIS 2.18. Variations in elevation, slope, and aspect within a region can lead to significant hydrological changes in the watershed, affecting both the speed and volume of runoff⁴⁷. Table 1 presents an overview of the topographical factors used in this study, including their descriptions, related equations, data sources, and value ranges.

Hydrological factors

Hydrological factors such as mean yearly precipitation, maximum 1-day precipitation, maximum 5-day precipitation, distance from rivers, and river density play crucial roles in flood modeling. Table 2 provides an

Variable [unite]	Description	Equation or data source [format sources]	Range (minimum–maximum)
Elevation [meter]	Elevation (Figure S1a) is a key factor in determining flood susceptibility, with lower elevation areas being more prone to flooding	NASA DEM (USGS Earth Explorer) [Raster (30 m)]	Max: 3633 Min: 530
Slope [degree]	Slope (Figure S2b) influences runoff and flow velocity, with flat areas being more prone to flood events due to slower water movement	Slope layer was produced using the DEM [Raster (30 m)]	Max: 76.81 Min: 0.60
Aspect [-]	Aspect (Figure S2c) analyzes the slope direction, influencing factors like water flow, precipitation reception, land cover patterns, and sunlight exposure	Aspect layer was produced using the DEM [Raster (30 m)]	10 classes
Topographic position index (TPI) [-]	TPI (Figure S2d) quantifies the elevation contrast between a pixel and its surrounding neighbors. It effectively identifies whether a point is situated on a ridge, valley, slope, or flat area	$TPI = \frac{E_{pixel}}{E_{neighbour}}$, E_{pixel} represents the elevation of the pixel, while $E_{neighbour}$ denotes the average value of the neighboring pixels [Raster (30 m)]	Max: 289 Min: -233
Stream power index (SPI) [-]	SPI (Figure S2e) serves as a gauge for a river's erosive power. Higher SPI values indicate stronger water flow capable of significant erosion, while areas with lower SPI values are more susceptible to receiving this erosive flow and are thus at increased risk of flooding ⁴⁸	$SPI = (a * tanb)$, 'a' denotes the specific drainage area, which is the area contributing water to a particular point in the river. 'b' represents the gradient or steepness of the land within that drainage area [Raster (30 m)]	Max: 2.00×10^8 Min: 10.00
Topographical roughness index (TRI) [-]	TRI (Figure S2f) affects flood risk, with smoother areas (lower TRI) having a higher likelihood of flooding ⁴⁹	$TRI = \sqrt{Abs(max^2 - min^2)}$ The minimum (min) value represents the lowest elevation, whereas the maximum (max) value indicates the highest elevation among the cells surrounding a specific point [Raster (30 m)]	Max: 0.90 Min: 0.11
Topographical wetness index (TWI) [-]	TWI (Figure S2g) quantifies the water storage capacity per pixel within the study area. A higher TWI value is associated with an elevated flood risk in that area ⁴³	$TWI = \frac{\ln(A_s)}{tan\beta}$, A_s indicates the specific catchment area (m ² /m ¹) and $\tan\beta$ indicates the slopes of the basin in degrees [Raster (30 m)]	Max: 26.15 Min: 1.45
Curvature [1/100 m]	Curvature (Figure S2h), which reflects the steepness of a slope, influences flood occurrence by impacting how water flows and accumulates on the land surface. Plan curvature (Fig. S2i) describes how the land surface curves in the horizontal plane, as seen from above, while profile curvature (Fig. S2j) describes how the land surface curves along a specific flow path, such as a river channel	Curvature layers was generated from the DEM [Raster (30 m)]	Max: 55.46 Min: -4.29
Plan curvature [1/100 m]			Max: 26.43 Min: -23.24
Profile curvature [1/100 m]			Max: 25.10 Min: -30.95
Convergence index (CI) [-]	CI (Figure S2k) summarizes terrain characteristics by measuring the average difference between the actual slope direction (aspect) and the theoretical direction of maximum water flow away from a peak ⁵⁰	$CI = \left(\frac{1}{8} \sum_{i=1}^8 \theta_i \right) - 90^\circ$, Here, θ represents the mean angle between the aspects of near by cells and the direction toward the central cell [Raster (30 m)]	Max: 100.00 Min: -99.77
Valley depth [meter]	Valley depth (Figure S2l), is essential for flood models to predict areas prone to inundation by analyzing the elevation difference between the valley floor and surrounding high points	Valley layer was produced using the DEM [Raster (30 m)]	Max: 830.00 Min: -118.00

Table 1. Topographical factors, detailing their definitions, relevant equations, data sources, and value ranges.

Variable [unite]	Description	Equation or data source [format sources]	Range (minimum–maximum)
Mean yearly precipitation [mm/year]	Rainfall (Figures S3–5a) is nearly always a prerequisite for flooding and is influenced by topographical and climatic factors	Daily precipitation data were collected from 4 synoptic stations and 18 rain gauge stations, then interpolated using the inverse distance method source: (IRAN Meteorological Organization, date: (1997 to 2020) [Vector (i.e., points)]	Max: 737 Min: 355
Maximum 1-day precipitation [mm]	Maximum precipitation in one day (Figure S3–5b)		Max: 192 Min: 68
Maximum 5-day precipitation [mm]	Maximum precipitation in 5 days (Figure S3–5c)		Max: 280 Min: 93
Distance from rivers [m]	The proximity to the river network (Figure S3d) significantly influences flood occurrence, with flood risk increasing closer to rivers and streams due to their role as natural drainage channels for floodwater ⁴⁰	Iran Water & Power Resources Development Company (IWPCO) [Raster (30 m)]	Max: 5077 Min: 0
River density [-]	River density (Figure S3e) is another significant hydrological factor; high river density indicates a more intricate network of streams and rivers. This network efficiently drains water from the surrounding area, potentially reducing the severity of floods		Max: 3.31 Min: 0.00

Table 2. Hydrological factors, detailing their definitions, relevant equations, data sources, and value ranges.

overview of the hydrological factors utilized in this study, along with their descriptions, associated equations, data sources, and ranges of values.

The average annual precipitation, calculated for the period 1997–2020, is higher in the central and northeastern parts, exceeding 671 mm/year, whereas the western and southeastern regions generally receive less, around 350 mm/year. This pattern is relatively consistent for maximum 1-day precipitation and maximum 5-day precipitation.

Environmental factors

Environmental factors, including the Normalized Difference Vegetation Index (NDVI), Normalized Difference Moisture Index (NDMI), Normalized Difference Water Index (NDWI), Normalized Difference Built-up Index (NDBI), LULC, lithology, and soil type, are essential components in flood modeling. Table 3 offers a summary of the environmental factors employed, including their descriptions, relevant equations, data sources, and value ranges.

Factor selection

Selecting the appropriate factors is crucial for identifying susceptible zones, as irrelevant factors introduce noise and diminish the model's effectiveness⁴⁰. In this research, two diagnostic methods were employed: the information gain ratio (IGR), which identifies the most impactful factors, and the variance inflation factor (VIF), which tests for multi-collinearity among the factors selected by the IGR method. The IGR test, a robust method for attribute selection, has been extensively utilized in prior studies^{39,43,54}. A high IGR indicates that the factor has a substantial effect on the target variable. The IGR can be calculated through Eqs. (1) to (4):

$$IG(D \cdot F) = \frac{Entropy(D) - Entropy(D \cdot F)}{SplitEntropy(D \cdot F)} \quad (1)$$

$$Entropy(D) = - \sum_{i=1}^2 \frac{(Y_i \cdot F)}{|D|} \log \frac{n(Y_i \cdot F)}{|D|} \quad (2)$$

$$Entropy(D \cdot F) = - \sum_{j=1}^m \frac{D_j}{|D|} Entropy(D) \quad (3)$$

$$SplitEntropy(D \cdot F) = - \sum_{j=1}^m \frac{D_j}{|D|} \log_2 \frac{D_j}{|D|} \quad (4)$$

Here, D represents the training set consisting of n input samples, and $n(Y_i \cdot D)$ denotes the number of input samples that fall into category Y_i (flood, no flood). Variance Inflation Factor (VIF) and tolerance are used to assess the presence of multicollinearity among independent factors. A $VIF > 5$ and $TOL < 0.1$ indicate multicollinearity issues⁵⁵. These factors can be calculated using Eqs. (5) and (6), respectively.

$$VIF_j = \frac{1}{1 - R_j^2} \quad (5)$$

$$T_j = 1 - R_j^2 \quad (6)$$

Here, R_j^2 represents the regression value of j , which is based on supplementary parameters in a given dataset⁵⁶.

Machine learning models

Extreme Gradient boosting

XGBoost, developed by Chen and Guestrin¹⁹, is a widely used machine learning algorithm recognized for its ability to overcome limitations related to computational speed and precision while requiring fewer resources for

Variable [unit]	Description	Equation or data source [format sources]	Range (minimum–maximum)
NDVI [-]	NDVI, depicted in Figure S4b, is a metric derived from remote sensing imagery that reflects vegetation density ⁴² . NDVI is calculated by analyzing reflectance in the red and near-infrared (NIR) regions of the electromagnetic spectrum ⁵¹	$\frac{(NIR(Band5) - Red(Band3))}{(NIR(Band5) + Red(Band3))}$ Red and near-infrared (NIR) represent the spectral reflectance in the respective red and near-infrared regions [Raster (30 m)]	Max: 0.84 Min: -0.43
NDMI [-]	NDMI, shown in Figure S4c, reflects plant moisture levels and can indicate flood risk, as denser vegetation generally helps mitigate flood severity ³²	$\frac{NIR(Band5) - SWIR(Band6)}{NIR(Band5) + SWIR(Band6)}$ NIR refers to the Near Infrared wavelength, and SWIR denotes the Short-Wave Infrared wavelength [Raster (30 m)]	Max: 0.54 Min: -0.46
NDWI [-]	NDWI, shown in Figure S4d, which identifies areas with high water content	$\frac{Green(Band3) - NIR(Band5)}{Green(Band3) + NIR(Band5)}$ [Raster (30 m)]	Max: 0.7 Min: -0.54
NDBI [-]	The NDBI is used to identify urban areas in satellite imagery by contrasting man-made structures with natural landscapes. It effectively distinguishes urban environments, helping assess impermeable surfaces	$\frac{SWIR(Band6) - NIR(Band5)}{SWIR(Band6) + NIR(Band5)}$ [Raster (30 m)]	Max: 0.46 Min: -0.54
LULC [-]	The primary factor influencing surface runoff, sediment quantity, and consequently, flooding in an area is LULC ⁵² , as shown in Figure S4a. Intensive vegetation typically reduces flood intensity ²⁴	This study utilizes Landsat-8 imagery from 2019 to extract LULC data [Raster (30 m)]	5 classes
Lithology [-]	Soil and Lithology directly influence water infiltration, affecting rainfall-runoff generation along with local weather conditions and erosion processes ^{49,53} . Higher infiltration rates reduce the likelihood of flooding. Figure S4f. and g depict the soil map and Lithology, respectively	IWPCO [Vector (i.e., polygons)]	25 classes
Soil type [-]			13 classes

Table 3. Environmental factors, detailing their definitions, relevant equations, data sources, and value ranges.

learning and prediction. It has been extensively utilized in various applications, including stock price prediction and human health risk assessment, demonstrating acceptable results¹⁹. As a tree-based ensemble model, XGBoost consists of multiple decision trees that work collaboratively to make predictions. Each tree assigns a score to a sample based on its characteristics. After training, the model forms several trees, each of which assigns a unique score to a sample's leaf nodes. The overall predicted value is then determined by summing the scores from all the trees⁵⁷.

$$(\phi) = \sum_{K=1}^K L(y_k \cdot \hat{y}_k) + \sum_{i=1}^N \Omega(g_i) \quad (7)$$

$$\Omega(g_i) = \gamma V + 0.5\lambda \|w\|^2 \quad (8)$$

According to Eq. (7), the objective function in XGBoost comprises two components: L , which represents the loss function between the actual and predicted values, and Ω , the penalty function. The regularization process in XGBoost includes the term Ω , which scales the penalty. Additionally, V denotes the number of leaves within a CART model, and w represents the vectors indicating the scores for these leaves. This regularization term effectively manages model complexity and helps prevent overfitting⁵⁷.

Support vector machine

The core idea behind SVM is to identify the best decision threshold in the data space that maximizes the separation between two classes while maintaining the maximum possible margin between the boundary and the data samples of both classes. SVM operates using kernel-based methods^{58–60}. The equation for SVM is expressed as follows:

$$f(x) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) K(x \cdot z) + b_i \quad (9)$$

The components of the SVM equation include the Lagrange multiplier, represented by $(\alpha_i - \alpha_i^*)$, the kernel function $K_{(x,z)}$ used within the multiplier, and the bias, denoted as b_i ⁶¹. In order to run the SVM models effectively, it is crucial to select or adjust the kernel function, which serves as the primary hyper parameter. The available kernel functions that can be utilized for SVM include polynomial, linear, sigmoid, and radial basis function (RBF)⁶¹. Further information on the SVM algorithm is available in Suthaharan⁶².

Random forest

RF is widely recognized for its effectiveness in addressing multi-classification and prediction tasks⁶³. RF assesses the importance of features by measuring how much the prediction error rises when the out-of-bag data for each feature is randomly shuffled while keeping others unchanged⁶⁴. In general, the randomness incorporated into RF algorithms helps mitigate overfitting by (i) constructing multiple trees, (ii) representing observations with replacements (bootstrapping), and (iii) randomly selecting subsets to determine the best splits at nodes⁶⁵. RF consists of a set of independent regression decision trees, each represented as $\{h(x, \theta_k), k \in 1, 2, \dots, k\}$ ^{63,66,67}

$$hx = \frac{1}{n} \sum h(x \cdot \theta_k) \quad (10)$$

The variable x represents the specific factor under examination, whereas θ_k is defined as an independent, identically distributed random variable. The variable n denotes the overall count of decision trees generated within the model⁶⁷. Detailed information about Random Forest can be found in Breiman³³.

Particle swarm optimization method

PSO is a population-driven algorithm that uses random variables to find optimal solutions⁶⁸. Inspired by the social behavior of birds in flocks, PSO involves particles that individually search for an optimal target variable. Each particle moves at its own velocity and continuously updates its position to find the best solution^{35,69}. The search process is influenced by each particle's best past experience as well as the best collective experience of all particles³⁵. The speed and position updates in this algorithm are calculated using Eqs. (11 and 12)⁷⁰:

$$V_{t+1} = W_t \times V_t + C_1 \times rand \times p_{best} - x_t + C_2 \times rand \times g_{best} - x_t \quad (11)$$

$$x_{t+1} = x_t + V_{t+1} \quad (12)$$

Here, W denotes the inertia weight, C_1 and C_2 are velocity constants, $rand()$ is a random function within the range $[0, 1]$, x_t represents the particle's current position, p_{best} is the particle's best individual location, and g_{best} is the optimal position discovered by the entire swarm. In our study, the objective function utilized within the PSO framework is maximizing of the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). The choice of AUC-ROC as the objective function is motivated by its effectiveness in evaluating the performance of binary classifiers, especially in scenarios with imbalanced datasets.

Validation and comparison

For evaluation purposes, criteria such as accuracy, precision, sensitivity, specificity, and the Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) were used. True Positive (TP) refers to the count of pixels accurately identified as a flood occurrence. False Positive (FP) is the count of pixels incorrectly identified as a flood event. True Negative (TN) denotes the count of pixels correctly identified as non-flood, while False Negative (FN) denotes the count of pixels incorrectly classified as non-flood. Accuracy, precision, sensitivity, and specificity are calculated using Eqs. (13) to (16):

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

$$Precision = \frac{TP}{TP + FP} \quad (14)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (15)$$

$$Specificity = \frac{TN}{TN + FP} \quad (16)$$

ROC curve graphically represents the detection capability of a binary classification system and is used to evaluate the test's sensitivity and specificity²⁴. This two-dimensional curve plots 1-specificity on the horizontal axis against sensitivity on the vertical axis. Sensitivity gauges the ability to accurately detect flood locations, while specificity assesses the ability to accurately identify non-flood locations (Plataridis and Mallios³⁸). According to Rahmati et al. (2018), the ROC curve is computed using Eq. (17):

$$AUC = \frac{\sum TP + \sum TN}{(P + N)} \quad (17)$$

Climate change

The Taylor diagram was utilized to find the best CMIP6 GCMs for precisely modeling future climate variables. This diagram integrates three statistical measures—correlation coefficient, centered root mean square error (CRMSE), and standard deviation—to assess the similarity between actual climate data and a model's results^{71,72}. In this study, an aggregate of 11 CMIP6 GCMs (see Table S1) was accessed to model monthly precipitation at the Khoramanad synoptic station during the baseline period from 1997 to 2015.

Regarding precipitation (Fig. 3), while the measured standard deviation was 26.72 mm, GCMs ranged from 5.1 mm to 32.01 mm in standard deviation. Except for TaiESM1, INM-CM4-8, and INM-CM5-0, most models depicted correlations above 0.74, with the majority having CRMSE values below approximately 18.1 mm. In summary, the analysis indicates that only eight models—FGOALS-g3, EC-Earth3-Veg-LR, EC-Earth3-Veg, EC-Earth3, GFDL-ESM4, MPI-ESM1-2-HR, ACCESS-ESM1-5, and ACCESS-CM2—correctly projected precipitation during the climate change period. These models were used for future climate studies.

Additionally, two different scenarios—SSP126 and SSP585—were used to consider a wide range of potential outcomes. To extract values from the GCMs for specific stations situated within the study area, the distances from each station and the center of the surrounding GCM pixels were calculated. The closest GCM pixel corresponding to each station was then selected as the matching target pixel.

The chosen GCMs were subject to bias correction using the well-established and efficient quantile mapping method. This method has proven to be highly effective in various hydrological and climate impact studies, highlighting its reliability and applicability^{73–79}. This process involves adjusting the distribution functions of simulated variables to match those measured, accomplished by calculating quantile associations through the conversion function. (Eq. 18)⁸⁰:

$$x^o = F_O^{-1} [F_m(x^m)] \quad (18)$$

where $F_m(x^m)$ is the cumulative distribution function (CDF) of measured data and F_O^{-1} is the inverse version of the CDF of measured data⁸⁰.

LULC

High-resolution satellite images are crucial for accurately describing LULC in the study area⁸¹. The current LULC status, categorized into rangeland, wetland, water, crops, and built areas, has been validated through visual interpretation using topographic maps and Google Earth images. LULC maps for the years 2000 and 2010 were created using the supervised classification in ENVI software, specifically employing the maximum likelihood algorithm. These maps are presented in Figures S5a and S5b.

The Land Change Modeler (LCM) module within TerrSet software was used to import the LULC maps from 2000 and 2010, along with DEM and slope maps of the research area, to predict future LULC changes. The LCM employs logistic regression and artificial neural networks (ANNs) to create transition potential maps. After developing these sub-models, the cellular Automata-Markov chain (CA-Markov) method was employed to predict the LULC map for 2020. This stochastic model generates a series of possible results, with the likelihood of each result influenced by neighboring outcomes, as indicated by a transition matrix^{81–83}. To evaluate the model's accuracy, the predicted 2020 map is assessed against the present map and verified using the Kappa coefficient, a statistical measure of agreement between two categorical datasets, calculated as outlined in Eq. (19):

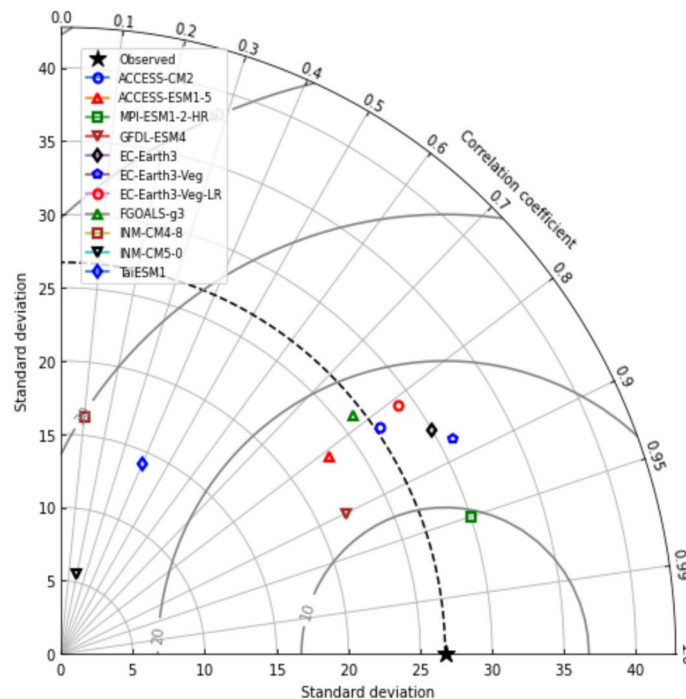


Fig. 3. Taylor diagram illustrating and comparing the abilities of 18 CMIP6 GCMs to replicate daily cumulative precipitation. The black dashed curved line indicates the standard deviation of the measured data.

$$k = \frac{P_o - P_e}{1 - P_e} \quad (19)$$

Here, P_o represents the observed agreement, and P_e denotes the expected agreement by chance. The model performance was found to be satisfactory with a Kappa coefficient of 0.92. Consequently, this model framework was used to predict LULC changes for the year 2050.

Results

Feature selection

IGR was used on the 24 initial factors to determine which ones have the most significant impact on flooding in the study area. The findings are shown in Fig. 4.

The results indicated that the distance from the river, with the highest IGR value of 0.45, had the greatest impact and is identified as the most significant factor, consistent with most previous studies^{42,52}. This is followed by the DEM with an IGR of 0.38, valley depth (0.28), TWI (0.22), and precipitation (0.18). These results align with findings from researchers such as Costache⁸⁴ and Talukdar et al.⁸⁵ regarding the fundamental influence of topography on floods^{84,85}. The NDVI has the lowest IGR value of 0.01, prompting the establishment of an IGR threshold of 0.15 to identify influential flood risk factors. Factors above this threshold significantly impact flood occurrence, enhancing model accuracy and efficiency. Out of 24 factors assessed, 10 with IGR values ≥ 0.15 were selected, including distance from the river, DEM, and slope. While precipitation is important for floods, topography proved more critical in this study, supporting findings by Plataridis and Mallios³⁸. Additionally, the soil type's IGR value of 0.06 may reflect that other factors, such as topography, LULC, and precipitation intensity, have a more direct and substantial impact on flood hazard. The influence of soil type, including its effects on infiltration and runoff rates, may be indirectly represented by these other variables such as LULC and TWI. Thus, while soil type is important, its relative impact on overall flood susceptibility is less pronounced compared to these more dominant factors.

VIF and Tolerances were used to assess the presence of multi-collinearity among 10 selected factors (Table 4).

As shown in Table 4, the VIF values for all factors are below 5, and the Tolerance values are all above 0.1. These results indicate that there is no significant multicollinearity among the selected factors. Also, the limitations of the methodology have been addressed and discussed in the Discussion section.

Machine learning hyper-parameter tuning

PSO was utilized to optimize the parameters of three machine learning algorithms: XGBoost, SVM, and RF. Prior to the optimization process, PSO parameters were configured with 100 evolutionary generations and a population size of 20, which are in line with the settings used in the study by Arabameri et al.⁵⁶. In the XGBoost model, essential parameters for performance include Max_depth, which influences tree complexity and overfitting risk; N_estimators, indicating the number of boosting iterations; and Learning_rate (eta), affecting step size during iterations. Lower Learning_rate values enhance robustness but require more rounds. Gamma sets the minimum

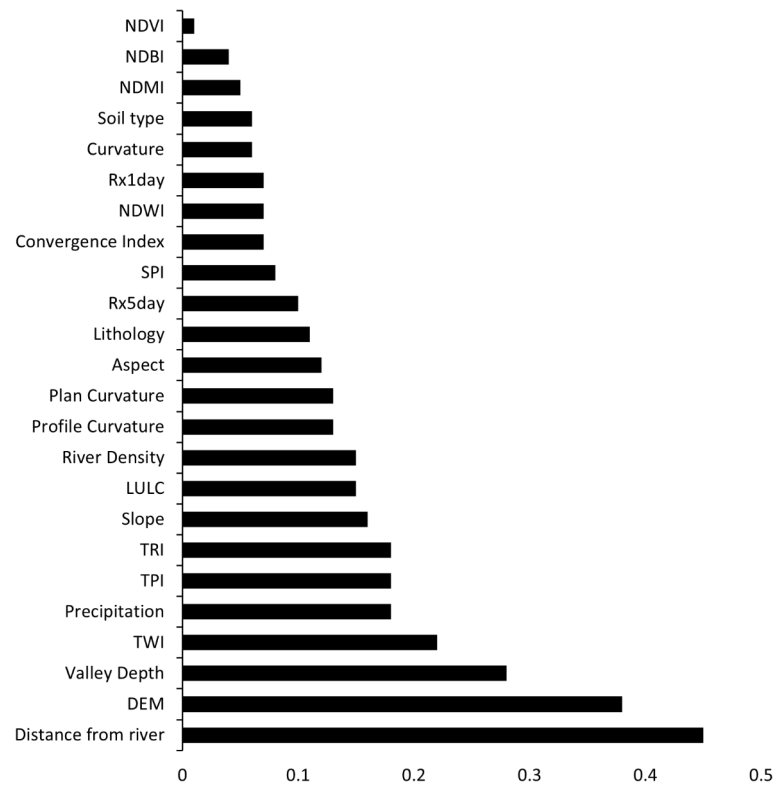


Fig. 4. Importance of predictive factors identified through the Information Gain Ratio method, showing key variables influencing flood susceptibility in the study area.

Variable name	VIF	Tolerance
Distance from river	2.352	0.425
DEM	2.998	0.334
Valley depth	2.366	0.423
TWI	1.905	0.525
Precipitation	1.685	0.593
TPI	1.442	0.693
TRI	1.389	0.720
Slope	2.021	0.495
LULC	1.298	0.771
River density	1.536	0.651

Table 4. VIF and tolerance values for flood condition factors, assessing multicollinearity among variables.

loss reduction for leaf node partitioning, promoting conservative models. Colsample_bytree and Subsample aid in generalization by determining the ratio of columns and training instances sampled, respectively, adding randomness to reduce overfitting. For SVM, optimizing kernel type and penalty factor (C) balances margin maximization and error reduction. In RF, Max_depth, N_estimators, Max_features, Min_samples_leaf, and Min_samples_split are vital for managing complexity and preventing overfitting, ultimately enhancing model robustness and predictive accuracy. The optimal values for XGBoost, SVM and RF are presented in Table 5.

Flood susceptibility maps

This section presents the results of using SVM-PSO, XGB-PSO, and RF-PSO algorithms to simulate flood susceptible areas in the study region for the reference period. By analyzing historical data on floods, precipitation, land use, and other relevant factors, these algorithms identify complex patterns that can be utilized to project future flood-prone areas. Flood probability maps were generated using XGBoost-PSO (Fig. 5a), SVM-PSO (Fig. 5b), and RF-PSO (Fig. 5c), with the results ranging from 0 to 1. A greater value suggests a higher chance of flooding. Employing the quantile technique, a widely recognized and reliable method for classifying results⁴², The flood probability maps were categorized into five groups: very low, low, moderate, high, and very high. In the XGBoost model (Fig. 5d), very low areas made up 17.06%, while moderate areas were 20.19%. The SVM model

Hyper-parameters	XGBoost		RF		SVM	
	Range [min, max]	Optimal value	Range [min, max]	Optimal value	Range [min, max]	Optimal value
Max_depth	[1, 20]	6.00	[1, 20]	5.00	–	–
N_estimators	[1, 300]	320.00	[1, 300]	251.00	–	–
Learning_rate	[0, 5]	0.20	–	–	–	–
gamma	[0, 10]	0.51	–	–	–	–
colsample_bytree	[0.5, 1]	0.65	–	–	–	–
subsample	[0.5, 1]	0.69	–	–	–	–
Max_features	–	–	1,5	1.00	–	–
Min_samples_leaf	–	–	2,10	8.00	–	–
Min_samples_split	–	–	2,10	3.00	–	–
Kernel	–	–	–	–	Linear, RBF, Sigmoid	Linear
C _{best}	–	–	–	–	[0.1, 100]	13.13

Table 5. Optimized parameters of machine learning models determined with PSO.

(Fig. 5f) had 7.7% in very low sensitivity, with moderate at 16.12%. The RF model (Fig. 5e) showed 20.19% for very low areas and 19.87% for moderate sensitivity. The flood-prone area classifications obtained from XGBoost, SVM, and Random Forest classifiers (Fig. 5g) show differences due to their unique learning methods. Variations arise from distinct variable selection and utilization techniques, affecting classification results and resulting in diverse maps⁸⁶. Analysis of machine-generated flood hazard maps, which represent the probability of a sample belonging to the “flood” class, with values ranging from 0 to 1, reveals that low-lying areas with gentle slopes and dense river networks are highly susceptible to flooding. These areas are further compromised by existing residential development, roads, and agricultural land use. This spatial correlation suggests a significant threat to human life and potential economic and social disruption in the event of a flood. Conversely, areas at higher elevations with steeper slopes and healthy or dense vegetation exhibit lower flood risk.

Accuracy evaluation of flood models

The assessment of three flood susceptibility maps utilized ROC curves and AUC values, indicating model efficacy in distinguishing flood from non-flood areas (Figure S6a and b). RF and XGBoost models excelled with AUCs over 0.99 for training data, showcasing excellent classification abilities. The SVM model performed well with an AUC of 0.96. On test data, RF led with an AUC of 0.98, followed by XGBoost (0.97) and SVM (0.95), confirming the link between high AUC values and accurate flood identification. Table 6 showcases the performance metrics of XGBoost, SVM, and RF in predicting flood-prone areas, evaluated on training and testing datasets. XGBoost and RF lead with the highest training accuracy at 0.96, though both show a slight decline to 0.92 during testing. RF excels in precision (0.95 training, 0.91 testing) and achieves the highest recall score of 0.98. Specificity further favors RF, scoring 0.94 in training and 0.91 in testing, reducing false positives. Overall, RF consistently delivers superior results across key metrics, making it the most reliable model for flood susceptibility predictions in this study.

The evaluation of models using ROC curves and AUC values indicates their strengths and weaknesses. All three models performed well (AUC > 0.95), with the RF model standing out due to its AUC of 0.98, demonstrating superior accuracy and precision for identifying flood-prone areas. XGBoost offered a solid alternative with an AUC of 0.97. However, the SVM model, while acceptable (AUC = 0.95), ranked lowest, with lower generalizability and effectiveness on complex datasets due to its linear decision boundary¹⁵. Therefore, based on the combined considerations of the statistical metrics the RF model is recommended for flood susceptibility forecasting and analysis within the context of this research, aligning with findings from other studies that recognized RF as a highly accurate model^{87,88}.

Future precipitation and LULC change projections

Eight global circulation models (GCMs) were utilized to downscale daily precipitation forecasts for the climate change period (2020–2050). The results indicated a decline in monthly precipitation trends relative to the baseline (1996–2015) under SSP126 (Fig. 6a) and SSP585 (Fig. 6b) scenarios, with an average reduction of 43.6 mm. The SSP126 scenario anticipates an average monthly precipitation of 42.38 mm, while the more severe SSP585 predicts a drop to 40.57 mm. Average annual precipitation maps reveal a maximum of 694 mm/year, below the baseline maximum of 737 mm/year. Minimum projected values (283 mm/year for SSP126 and 273 mm/year for SSP585) also fall short of the baseline minimum (355 mm/year). We employed annual average precipitation to develop future flood susceptibility models, which is in line with studies by Roy et al.⁸⁸, Avand et al.³¹, and Mahdizadeh-Gharakhanlou and Perez³².

This study investigated LULC changes in the studied area over a 20-year period (2000–2020) using Landsat satellite imagery from 2000, 2010, and 2020. LULC maps were generated for each time period (Figure S7a–c). These maps were then analyzed to identify and quantify changes in LULC across various classes. The analysis focused on both the magnitude and direction of change (expansion or reduction) for each land use class, including transitions between classes. The area of each LULC class was calculated from the classified maps. This allowed for a clear visualization of changes in each class over the 20-year period. By comparing these maps, the percentage change for each class was determined (Table 7). In the year 2000, rangeland represented the

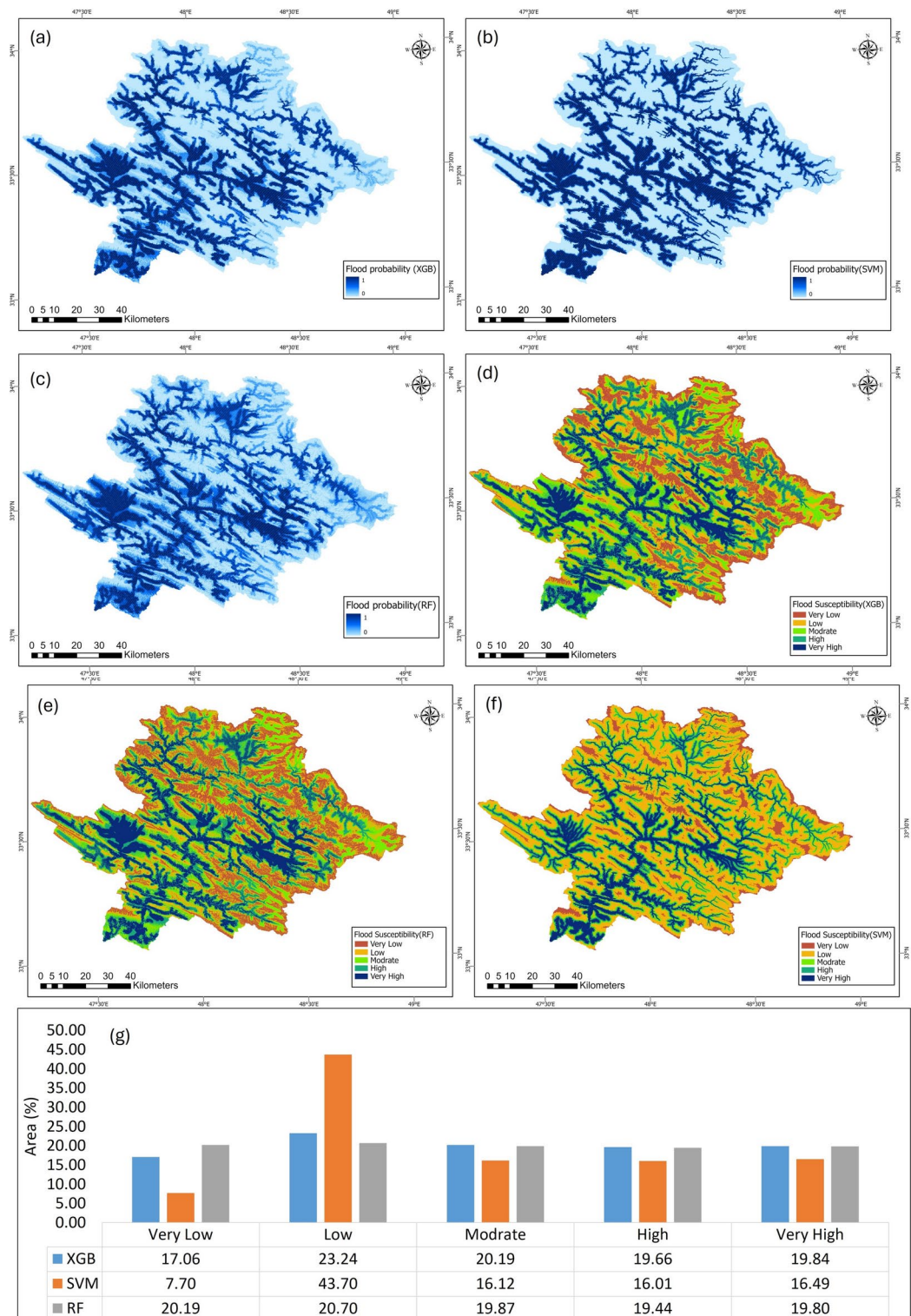


Fig. 5. Flood probability maps generated by XGBoost (a), RF (b), and SVM (c), alongside flood susceptibility maps produced by XGBoost (d), RF (e), and SVM (f). Graph (g) details the proportion of areas categorized by each model [the maps were generated by QGIS 2.18 (<https://qgis.org/>)].

dominant LULC with the largest area (7233 km²), while water bodies exhibited the smallest area (1.95 km²). Over the following two decades, pasture area decreased by 2.15%, reaching 7027 km² in 2020. Conversely, crops increased by 0.73%, expanding from 2168 km² to 2238 km² in 2020. Notably, the area of built area significant doubling from 132 km² (1.39%) in 2000 to 265.88 km² (2.78%) in 2020.

Metrics	Train data			Test data		
	XGBoost	SVM	RF	XGBoost	SVM	RF
True positive	352	347	354	124	124	129
False positive	20	28	18	19	19	14
True negative	301	293	303	144	136	139
False negative	10	15	8	7	15	12
Accuracy	0.96	0.94	0.96	0.92	0.87	0.92
Precision	0.95	0.93	0.95	0.9	0.87	0.91
Recall	0.97	0.96	0.98	0.95	0.89	0.92
Specificity	0.94	0.91	0.94	0.88	0.88	0.91

Table 6. Performance metrics for machine learning models applied to training and testing datasets, assessing flood susceptibility predictions.

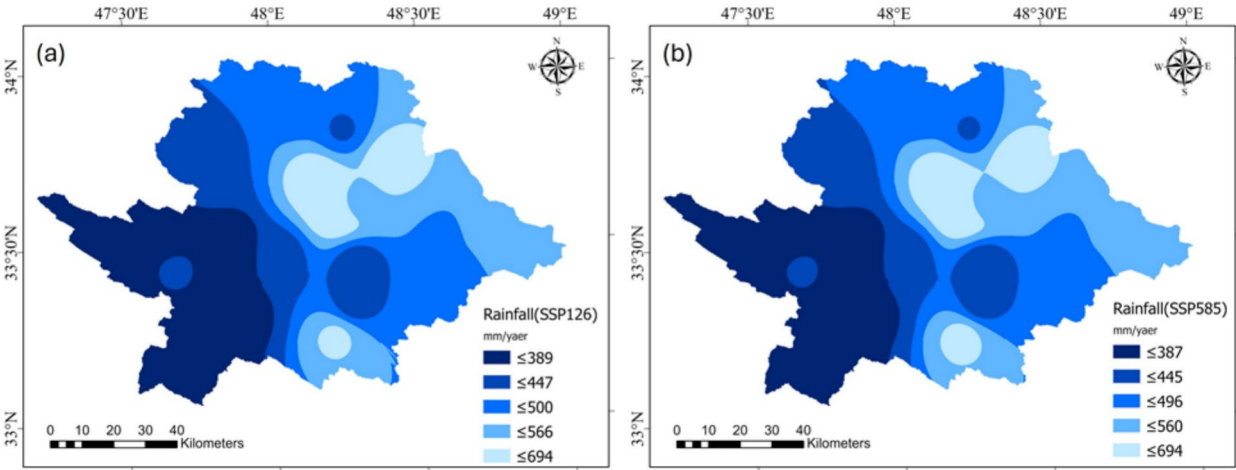


Fig. 6. Spatial distribution of average annual precipitation projected for 2020–2050 under SSP126 (a) and SSP585 (b) climate scenarios [the maps were generated by QGIS 2.18 (<https://qgis.org/>)].

Year	2000		2010		2020	
LULC	Area (km ²)	Area (%)	Area (km ²)	Area (%)	Area (km ²)	Area (%)
Rangeland	7233.60	75.79	7142.43	74.84	7027.60	73.64
Wetland	6.77	0.069	6.60	0.07	4.70	0.05
Water	1.95	0.02	2.31	0.02	6.93	0.07
Crops	2168.79	22.72	2205.62	23.11	2238.06	23.45
Built Area	132.06	1.39	186.21	1.95	265.88	2.78

Table 7. Temporal distribution of LULC classes (km²) for 2000, 2010, and 2020, highlighting major LULC changes over time.

The projected LULC map for 2050, derived from the Cellular Automata-Markov chain (CA-Markov) model (Fig. 7a), shows a high accuracy of 92% when compared with the actual 2020 LULC map from Landsat. The model forecasts that rangeland will decrease to approximately 5936.07 km² (62% of the basin) by 2050, while built areas are expected to rise to 859.30 km² (9% of the basin). From 2020 to 2050 (Fig. 7b), rangeland is projected to lose 1091.53 km², indicating significant contraction. Wetlands will see slight stability with a 10.42 km² increase, and water bodies will gradually expand, particularly from 2010 to 2020 (4.61 km²). Cropland and built areas will grow notably post-2020, with increases of 476.3 km² and 593.42 km², respectively.

Projection of flood-prone areas in future periods

This study investigated future flood susceptibility in the study area by integrating LULC projections and climate scenarios. The RF model was employed for this goal. First, LULC for the year 2050 was projected using the methods described previously. Next, average annual precipitation for 2020–2050 was obtained under two contrasting climate scenarios: SSP126 and pessimistic SSP585. These scenarios represent a range of potential

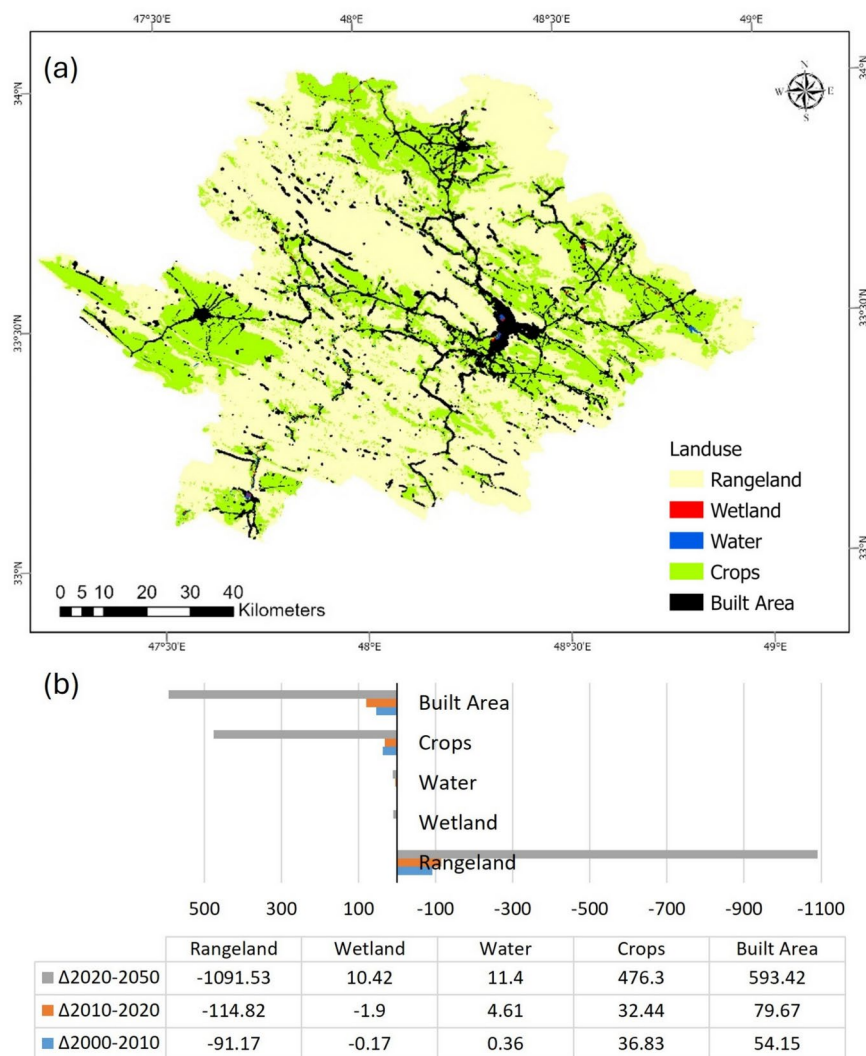


Fig. 7. Spatial maps of LULC projections (a) and quantified changes across different time periods (b), illustrating significant transitions in LULC patterns [the map was generated by QGIS 2.18 (<https://qgis.org/>)].

future climate conditions. The predicted LULC map for 2050 and the corresponding future precipitation data (from both scenarios) were then used as inputs for the RF model. The model was trained to identify susceptible areas to flooding based on these combined factors for SSP126 (Fig. 8a) and SSP585 (Fig. 8b). Finally, the model outputs were analyzed to assess changes in flood risk under the different climate scenarios, as shown in Table 8.

The percentage of flood-prone areas with different capacities in SSP126 and SSP585 scenarios is calculated as follows: The percentage of areas with very low potential for flooding in the base period is 20.19% and will decrease to 19.93% and 19.96% in the SSP126 and SSP 585 scenarios, respectively. The percentage of areas with low potential will decrease from 20.70% in the basic period to 20.67% in SSP126 and 20.64% in SSP585. The percentage of regions with average potential was 19.87% in the base period and will change to 19.90 in SSP126 and 19.89 in SSP 585 in both scenarios in the future. The percentage of areas with high potential for flooding will increase from 19.44% in the base period to 19.83% in SSP126 and 19.82% in SSP585, and finally, the percentage of areas with very high potential for flooding will decrease from 19.80% in the base period will increase to 19.68% in SSP126 and 19.69% in SSP585.

Discussion

Flooding represents a major global natural disaster, particularly affecting regions susceptible to urbanization and climate change. This study explores future flood susceptibility in the Kashkan watershed of Iran, utilizing machine learning models that integrate climate change projections and land use/land cover (LULC) changes. Our results highlight the critical relationship between human activity and environmental factors that elevate flood risks. Notably, the RF model yielded the best predictive performance, achieving a 92% accuracy and an AUC of 0.98, showcasing the effectiveness of ML models for environmental assessments. Key determinants of flood susceptibility included proximity to rivers, DEM, precipitation, and LULC. The increase in built-up areas from 2000 to 2020 is attributed to profound population growth in Lorestan province, where the population

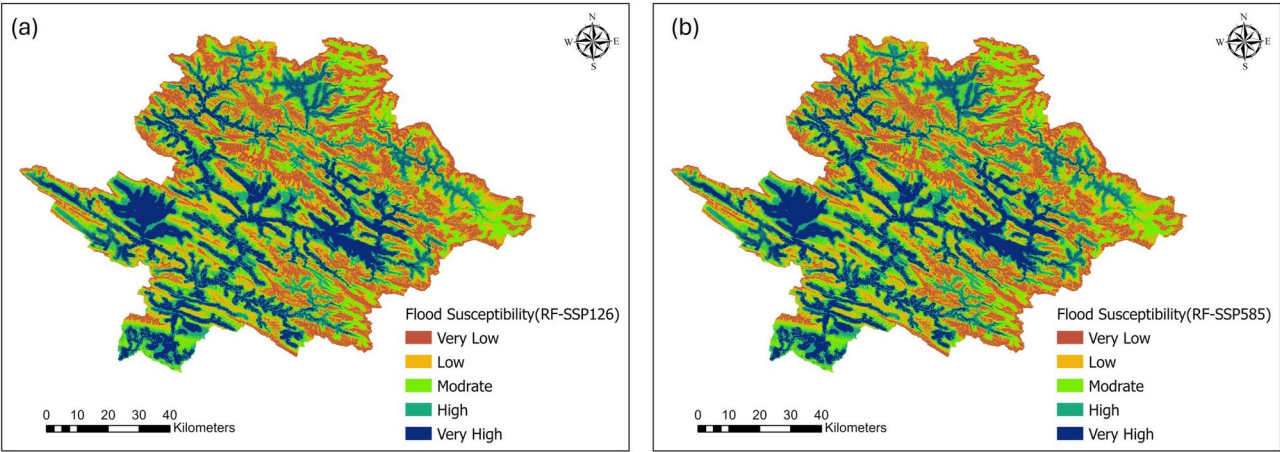


Fig. 8. Classification of future flood-prone areas for 2050 under SSP126 (a) and SSP585 (b) scenarios, highlighting shifts in susceptibility driven by projected climate and LULC changes [the maps were generated by QGIS 2.18 (<https://qgis.org/>)].

Flood susceptibility	Base period	SSP126	SSP585
Very low	20.19	19.93	19.96
Low	20.70	20.67	20.64
Moderate	19.87	19.90	19.89
High	19.44	19.83	19.82
Very high	19.80	19.68	19.69

Table 8. Comparative percentages of flood susceptibility areas for the base period and future scenarios, showing changes due to climate and LULC transitions.

surged from approximately 686,000 in 1976 to 1,764,000 in 2016. This demographic change has intensified the demand for land for housing and urban infrastructure, especially in Khorramabad, the provincial capital at the center of the watershed. Projections for 2050 indicate a notable expansion in moderate and high-risk flood zones under SSP585 and SSP126 scenarios, signaling heightened flood risks due to climate and land use shifts. To effectively mitigate these risks, comprehensive planning and sustainable management of resources are essential, echoing findings from recent literature on climate variability and LULC transformations.

This research offers several novel contributions to the field of flood risk management. By integrating CMIP6 climate projections with LULC changes, the study significantly enhances the predictive accuracy of flood susceptibility assessments. The application of PSO to refine machine learning models further strengthens their robustness and reliability. Additionally, the incorporation of dynamic LULC simulations, as emphasized in previous studies, provides a comprehensive framework for evaluating future flood risks, particularly in climate-vulnerable regions such as the Middle East. The projected expansion of high-risk flood zones emphasizes the pressing need for sustainable urban planning and effective water resource management strategies. These insights are critical for policymakers tasked with mitigating flood risks, especially in rapidly urbanizing areas. Our findings not only corroborate but also extend existing research on flood susceptibility modeling. In alignment with Madhuri et al.³⁰ and Avand et al.³¹, our results validate the effectiveness of machine learning techniques in identifying flood-prone regions, particularly in areas with limited data availability^{30,31}. Moreover, the methodological advancements, particularly the integration of optimization algorithms, underscore the importance of multidimensional approaches in flood risk analysis. Furthermore, the role of urban expansion in exacerbating flood risks, as discussed by Recanatesi and Petroselli³, provides additional context to our findings, reinforcing the significance of addressing urban growth in flood risk management strategies.

While this study offers valuable insights into flood risk management, several limitations should be acknowledged. First, the reliance on historical flood data introduces the potential for temporal biases, as past flood events may not fully represent future scenarios under changing climate conditions. Although the CMIP6 climate models provide advanced and comprehensive projections, the inherent uncertainties in climate scenarios—such as regional variability, model assumptions, and future emissions pathways—may affect the reliability and accuracy of the flood susceptibility maps generated in this study. Additionally, the exclusion of key socioeconomic indicators, such as population density and infrastructure resilience, limits the comprehensiveness of the analysis. These factors are crucial for understanding how human activities and socio-economic dynamics influence flood risks. Incorporating such data could significantly enhance the robustness of flood risk assessments, especially in rapidly urbanizing or socioeconomically vulnerable regions.

Conclusion

Flood susceptibility mapping remains a pivotal aspect of managing natural hazards, emphasizing the importance of timely and accurate models to mitigate flood risks. This study introduced advanced machine learning approaches to identify and project current and future flood-prone areas in the Kashkan catchment, incorporating topographical, environmental, and hydrological factors. The analysis is constrained by the spatial and temporal resolution of input datasets, including DEM and rainfall data, which may affect the precision of flood risk mapping. Additionally, the focus on environmental and geophysical parameters excludes key socio-economic factors, such as population growth and infrastructure resilience, limiting the comprehensiveness of the assessment. Furthermore, the reliance on historical data for model calibration and validation introduces uncertainties, particularly when projecting long-term impacts under diverse climate scenarios. To enhance the accuracy and comprehensiveness of flood-prone area assessments, the following recommendations for future studies are suggested:

- Calculate flood risk maps by using machine learning methods with other criteria to represent exposure and vulnerability indices, such as GDP index, household income index, population, environmental indicators, and indicators related to flood-vulnerable economic activities
- Predict factors affecting floods beyond LULC and precipitation for the climate change period

Data availability

Correspondence and requests for materials should be addressed to H.K.

Received: 20 November 2024; Accepted: 1 April 2025

Published online: 11 April 2025

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H.K.: Methodology, Conceptualization, Data Curation, Software, Visualization, Validation, Investigation, Writing—Original Draft, Review & Editing; F.N.S.: Project administration, Methodology, Advising, Review & Editing; A.N.D.: Methodology, Conceptualization, Data Curation; E.Z.: Methodology, Advising, Writing—Original Draft, Review & Editing.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-025-97008-0>.

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